

ICE CAPSULE CHARGING & DISCHARGING PERFORMANCE TEST REPORT

1. Objectives

To evaluate the charging and discharging performance of the Ice Capsule by monitoring its temperature profile under controlled water-glycol bath conditions.

2. Test Sample

Product: Ice Capsule / HDPE Cryogel Ball

3. Test Conditions

Parameters	Charging Condition	Discharging Condition
Medium	Water-Glycol Solution	Water-Glycol Solution
Bath Temperature	-7 °C	7 °C
Initial Ball Temperature	+5 °C	-7 °C

4. Test Procedure

4.1 Charging Test

The Ice Capsule was conditioned at +5 °C and immersed in a water-glycol bath maintained at -7 °C. The temperature of the capsule was continuously monitored until thermal equilibrium was achieved.

4.2 Discharging Test

The charged Ice Capsule at -7 °C was transferred to a water-glycol bath maintained at 7 °C. The temperature was continuously monitored until thermal equilibrium was achieved.

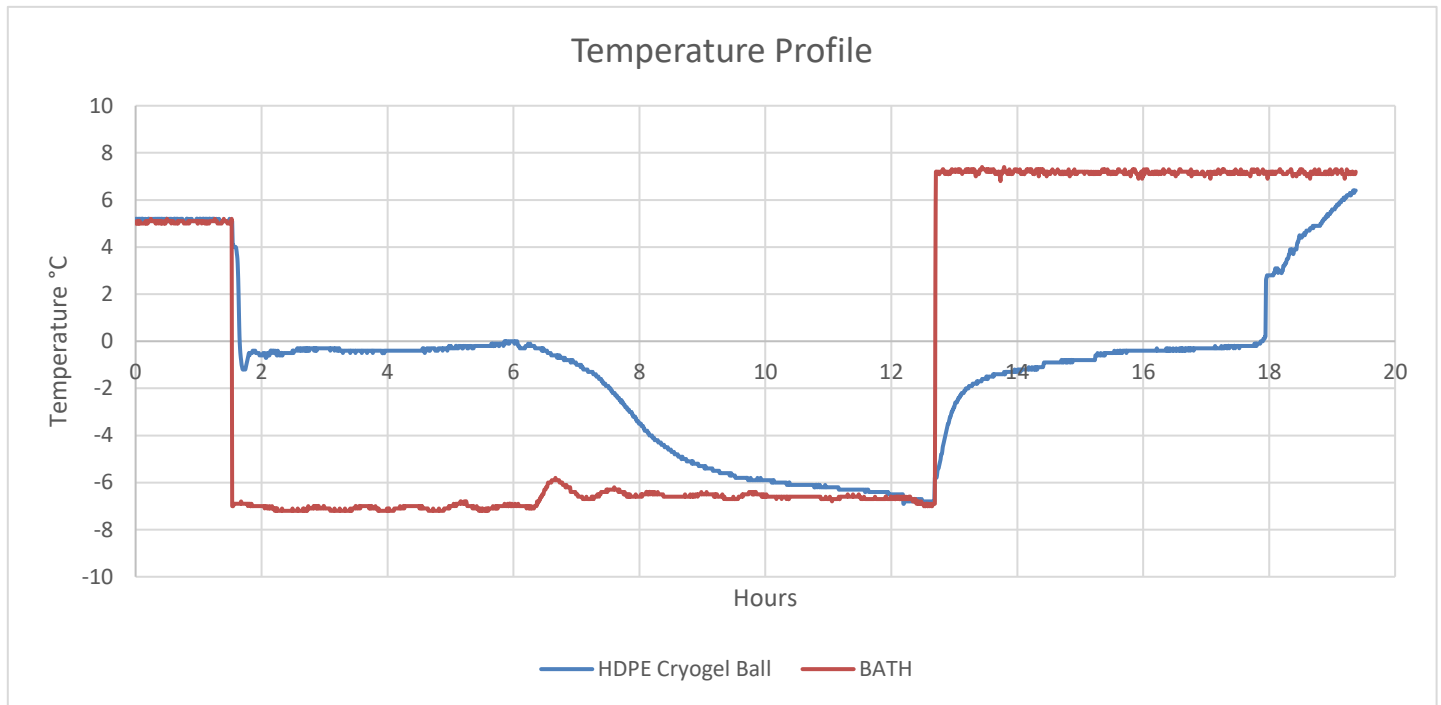
5. Results

- The total charging time recorded was **10.4 hours** from 5 °C to -7 °C.
- The total discharging time recorded was **6.6 hours** from -7 °C to 7 °C.
- No leakage, deformation, or physical damage was observed during testing.

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6. Temperature Profile



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